



**Tertiary Infotech Academy Pte Ltd**

UEN: 201200696W

# **LESSON PLAN**

For

## **Autodesk Certified Professional (ACP) - AutoCAD Design and Drafting**

TGS Ref No: TGS-2023036004

Conducted by

**Tertiary Infotech Academy Pte Ltd**

UEN: 201200696W

**Version 8.0**

## DOCUMENT VERSION CONTROL RECORD

| Version Number | Effective Date of Release | Summary of Included Changes                                                                                                                  | Author                    |
|----------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| 1.0            | 14 June 2023              | First version                                                                                                                                | Tertiary Infotech Academy |
| 2.0            | 4 September 2024          | Second version                                                                                                                               | Tertiary Infotech Academy |
| 3.0            |                           | New format                                                                                                                                   | Tertiary Infotech Academy |
| 4.0            | 17 October 2025           | Updated company name                                                                                                                         | Tertiary Infotech Academy |
| 8.0            | 22 August 2026            | Rebuilt 32-hour schedule aligned to v8 slides, ten AutoCAD labs, detailed Learner Guide, recovered PP dataset and revised WA/PP instruments. | Tertiary Infotech Academy |

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## Course Information

**Course title:** Autodesk Certified Professional (ACP) - AutoCAD Design and Drafting

**Course code:** TGS-2023036004

**TSC:** Design Sketching · BEV-DES-5026-1.1-1

**Duration:** 4 days · 32 contact hours, including 3 hours of summative assessment

**Mode:** Instructor-led concept briefing, AutoCAD demonstration, individual lab practice, evidence review and assessment

**Trainer:** Dr Alfred Ang

## Learning Outcomes

- LO1: Define the application of emerging design visualization standards in AutoCAD art direction and sketching.
- LO2: Develop the organisation's drawing and sketching capabilities using methods for revising 2D and 3D technical drawings.
- LO3: Develop organisation-specific drawing standards that use storytelling and annotation to engage stakeholders.
- LO4: Lead the integration of emerging drawing technologies and suitable methods in alignment with organisational growth.
- LO5: Inspire stakeholders to explore traditional and contemporary approaches to drawing and sketching.

## Assessment

- Written Assessment (WA-SAQ) — 4 open-ended questions covering K1–K4, 60 minutes, open book.
- Practical Performance (PP) — 5 hands-on tasks covering A1–A5, 120 minutes, open book.
- Learners must demonstrate every required K and A criterion to achieve Competent (C).

## Course Schedule

### Day 1 - Standards, drawing management and advanced geometry

| Time      | Duration | Block              | Content / activity                                                                                          | Resources   |
|-----------|----------|--------------------|-------------------------------------------------------------------------------------------------------------|-------------|
| 9:00-9:30 | 30 min   | <b>Orientation</b> | Welcome, digital attendance, trainer/learner introductions, learning outcomes, TSC and assessment briefing. | Slides 2-16 |

| Time        | Duration | Block           | Content / activity                                                                                      | Resources                   |
|-------------|----------|-----------------|---------------------------------------------------------------------------------------------------------|-----------------------------|
| 9:30-10:30  | 60 min   | <b>Domain 1</b> | Standards, project setup, units, scale, UCS and drawing governance; concept briefing and demonstration. | Slides 17-21                |
| 10:30-10:45 | 15 min   | <b>Break</b>    | Tea break.                                                                                              | -                           |
| 10:45-12:15 | 90 min   | <b>Lab 1</b>    | Drawing Standards and Project Setup; individual AutoCAD practice and evidence review.                   | Slide 26 · Lab 1 · LG Lab 1 |
| 12:15-13:15 | 60 min   | <b>Lunch</b>    | Lunch break.                                                                                            | -                           |
| 13:15-14:15 | 60 min   | <b>Domain 1</b> | Layer governance, layer states, filters, viewport overrides, selection and drawing health.              | Slides 22-25                |
| 14:15-15:15 | 60 min   | <b>Lab 2</b>    | Layers, Object Management and Revision Control.                                                         | Slide 27 · Lab 2 · LG Lab 2 |
| 15:15-15:30 | 15 min   | <b>Break</b>    | Tea break.                                                                                              | -                           |
| 15:30-17:15 | 105 min  | <b>Domain 3</b> | Advanced drawing objects, object-type selection and the precision/editing system.                       | Slides 39-44                |
| 17:15-18:00 | 45 min   | <b>Review</b>   | Measured-geometry critique, Day 1 retrieval practice and PM attendance.                                 | Slides 41-44                |

**Day total:** 480 contact minutes (8 hours), excluding lunch and including scheduled tea breaks.

## Day 2 - Precision editing, reusable content and annotation

| Time        | Duration | Block           | Content / activity                                                                                | Resources                   |
|-------------|----------|-----------------|---------------------------------------------------------------------------------------------------|-----------------------------|
| 9:00-9:15   | 15 min   | <b>Review</b>   | AM attendance and Day 1 retrieval practice.                                                       | Selected Day 1 slides       |
| 9:15-10:15  | 60 min   | <b>Domain 2</b> | Stakeholder reading order, annotation styles, dimensions, hatches, revision and annotative scale. | Slides 28-35                |
| 10:15-10:30 | 15 min   | <b>Break</b>    | Tea break.                                                                                        | -                           |
| 10:30-11:45 | 75 min   | <b>Lab 3</b>    | Annotation Standards and Stakeholder Story.                                                       | Slide 37 · Lab 3 · LG Lab 3 |
| 11:45-12:30 | 45 min   | <b>Lab 4</b>    | Dimensions, Multileaders and Data Tables.                                                         | Slide 38 · Lab 4 · LG Lab 4 |
| 12:30-13:30 | 60 min   | <b>Lunch</b>    | Lunch break.                                                                                      | -                           |
| 13:30-14:30 | 60 min   | <b>Domain 3</b> | Precision editing, arrays, grips, regions, isometric drafting, blocks and groups.                 | Slides 43-48                |

| Time        | Duration | Block         | Content / activity                                  | Resources                   |
|-------------|----------|---------------|-----------------------------------------------------|-----------------------------|
| 14:30-15:45 | 75 min   | <b>Lab 5</b>  | Advanced Geometry and Precision Editing.            | Slide 50 · Lab 5 · LG Lab 5 |
| 15:45-16:00 | 15 min   | <b>Break</b>  | Tea break.                                          | -                           |
| 16:00-17:15 | 75 min   | <b>Lab 6</b>  | Blocks, Attributes and Data Extraction.             | Slide 51 · Lab 6 · LG Lab 6 |
| 17:15-18:00 | 45 min   | <b>Review</b> | Reuse/data critique, Day 2 recap and PM attendance. | Slides 49-51                |

**Day total:** 480 contact minutes (8 hours), excluding lunch and including scheduled tea breaks.

### Day 3 - Layouts, publishing and collaboration

| Time        | Duration | Block           | Content / activity                                                                  | Resources                   |
|-------------|----------|-----------------|-------------------------------------------------------------------------------------|-----------------------------|
| 9:00-9:15   | 15 min   | <b>Review</b>   | AM attendance and Day 2 retrieval practice.                                         | Selected Day 2 slides       |
| 9:15-10:15  | 60 min   | <b>Domain 4</b> | Model/paper space, layouts, page setups, viewports, scales and output quality.      | Slides 52-57                |
| 10:15-10:30 | 15 min   | <b>Break</b>    | Tea break.                                                                          | -                           |
| 10:30-12:00 | 90 min   | <b>Lab 7</b>    | Layouts, Viewports and Page Setups using the recovered sink dataset.                | Slide 60 · Lab 7 · LG Lab 7 |
| 12:00-13:00 | 60 min   | <b>Lab 8</b>    | Publish, Plot and eTransmit.                                                        | Slide 61 · Lab 8 · LG Lab 8 |
| 13:00-14:00 | 60 min   | <b>Lunch</b>    | Lunch break.                                                                        | -                           |
| 14:00-15:00 | 60 min   | <b>Domain 5</b> | Xrefs, underlays, paths, comparison and emerging collaboration workflows.           | Slides 62-69                |
| 15:00-15:15 | 15 min   | <b>Break</b>    | Tea break.                                                                          | -                           |
| 15:15-16:30 | 75 min   | <b>Lab 9</b>    | Xrefs, Underlays and Drawing Comparison.                                            | Slide 71 · Lab 9 · LG Lab 9 |
| 16:30-17:30 | 60 min   | <b>Capstone</b> | Brief the integrated sink-documentation problem and technology-selection scorecard. | Slides 70-72                |
| 17:30-18:00 | 30 min   | <b>Review</b>   | Dependency/path check, capstone plan, Day 3 recap and PM attendance.                | Lab 9-10 briefs             |

**Day total:** 480 contact minutes (8 hours), excluding lunch and including scheduled tea breaks.

### Day 4 - Integrated capstone, evidence review and assessment

| Time      | Duration | Block         | Content / activity                          | Resources    |
|-----------|----------|---------------|---------------------------------------------|--------------|
| 9:00-9:15 | 15 min   | <b>Review</b> | AM attendance and capstone readiness check. | Lab 10 brief |

| Time        | Duration | Block             | Content / activity                                                                                             | Resources                                    |
|-------------|----------|-------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| 9:15-10:15  | 60 min   | <b>Lab 10</b>     | Integrated Sink Documentation<br>Capstone - planning, standards and sheet setup.                               | Slide 72 · Lab 10 · LG<br>Lab 10             |
| 10:15-10:30 | 15 min   | <b>Break</b>      | Tea break.                                                                                                     | -                                            |
| 10:30-12:30 | 120 min  | <b>Lab 10</b>     | Capstone production - viewports, title blocks, component table, revision communication, publish and eTransmit. | Lab 10 folder · recovered DWGs · detailed LG |
| 12:30-13:30 | 60 min   | <b>Lunch</b>      | Lunch break.                                                                                                   | -                                            |
| 13:30-14:30 | 60 min   | <b>Review</b>     | Capstone evidence review, peer explanation and course summary.                                                 | Slides 74-75                                 |
| 14:30-14:45 | 15 min   | <b>Break</b>      | Tea break.                                                                                                     | -                                            |
| 14:45-15:00 | 15 min   | <b>Assessment</b> | Final assessment briefing and assessment digital attendance.                                                   | Slides 15-16                                 |
| 15:00-16:00 | 60 min   | <b>WA-SAQ</b>     | Four open-ended questions covering K1-K4.                                                                      | WA question paper                            |
| 16:00-18:00 | 120 min  | <b>PP</b>         | Five practical tasks covering A1-A5 using the sink-layout dataset; feedback and sign-off.                      | PP question paper · AutoCAD dataset          |

**Day total:** 480 contact minutes (8 hours), excluding lunch and including scheduled tea breaks.

## Alignment Map

- Domain 1 and Labs 1-2 -> LO1-LO2 · K1-K2 · A1-A2.
- Domain 2 and Labs 3-4 -> LO3 · K3 · A3.
- Domain 3 and Labs 5-6 -> LO1-LO4 · K1, K2, K4 · A1, A2, A4.
- Domain 4 and Labs 7-8 -> LO3-LO4 · K3, K4 · A1-A4.
- Domain 5 and Labs 9-10 -> LO1-LO5 · K1-K4 · A1-A5.
- WA questions 1-4 -> K1-K4; PP tasks 1-5 -> A1-A5.